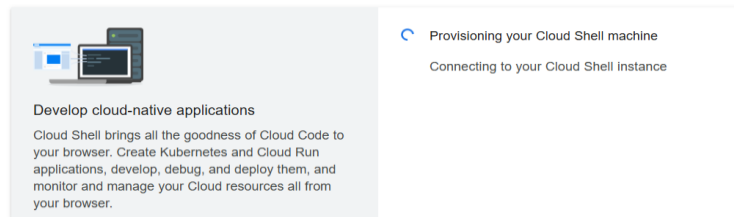
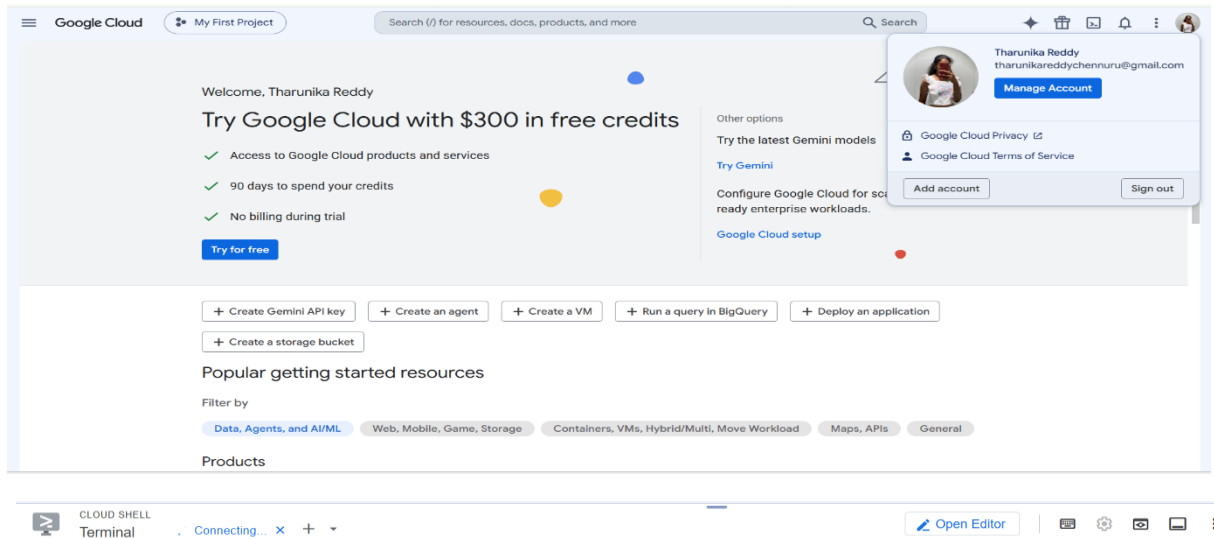


GOOGLE CLOUD LAB

1a. Execute a Python Program in Google Cloud Shell

Aim: Run a Python application using a cloud-based environment.



Google CloudMy First ProjectSearch (/) for resources, docs, products, and moreSearch

CLOUD SHELLTerminal(cogent-tine-479708-f6) x +

Open Editor

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to cogent-tine-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6) \$ nano hello.py

Google CloudMy First ProjectSearch (/) for resources, docs, products, and moreSearch

CLOUD SHELLTerminal(cogent-tine-479708-f6) x +

Open Editor

GNU nano 2.2.8hello.py
print("Hello From Google Cloud")
print("Welcome to Cloud Computing")

HelpExitWrite OutRead FileWhere IsReplaceCutPasteExecuteJustifyRead 2 linesLocationGo To LineUndoRedoSet MarkCopyTo BracketWhere WasPreviousNextBackForward

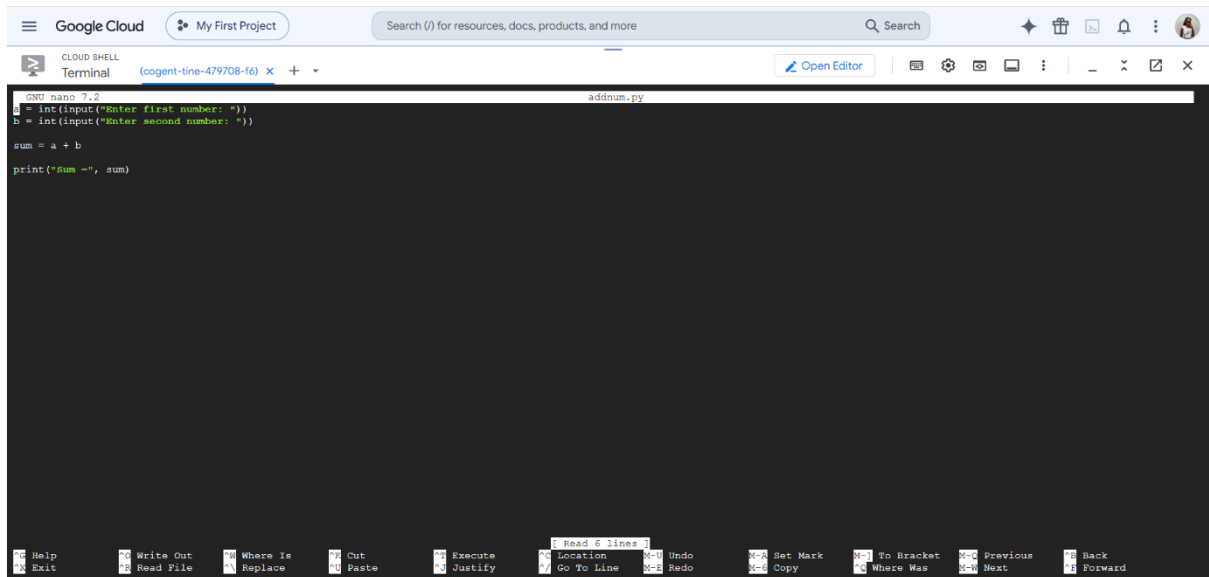
Google CloudMy First ProjectSearch (/) for resources, docs, products, and moreSearch

CLOUD SHELLTerminal(cogent-tine-479708-f6) x +

Open Editor

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to cogent-tine-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6) \$ nano hello.py
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6) \$ python3 hello.py
Hello From Google Cloud
Welcome to Cloud Computing
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6) \$

1.b Execute a Python Program in Google Cloud Shell for Addition of Two Numbers



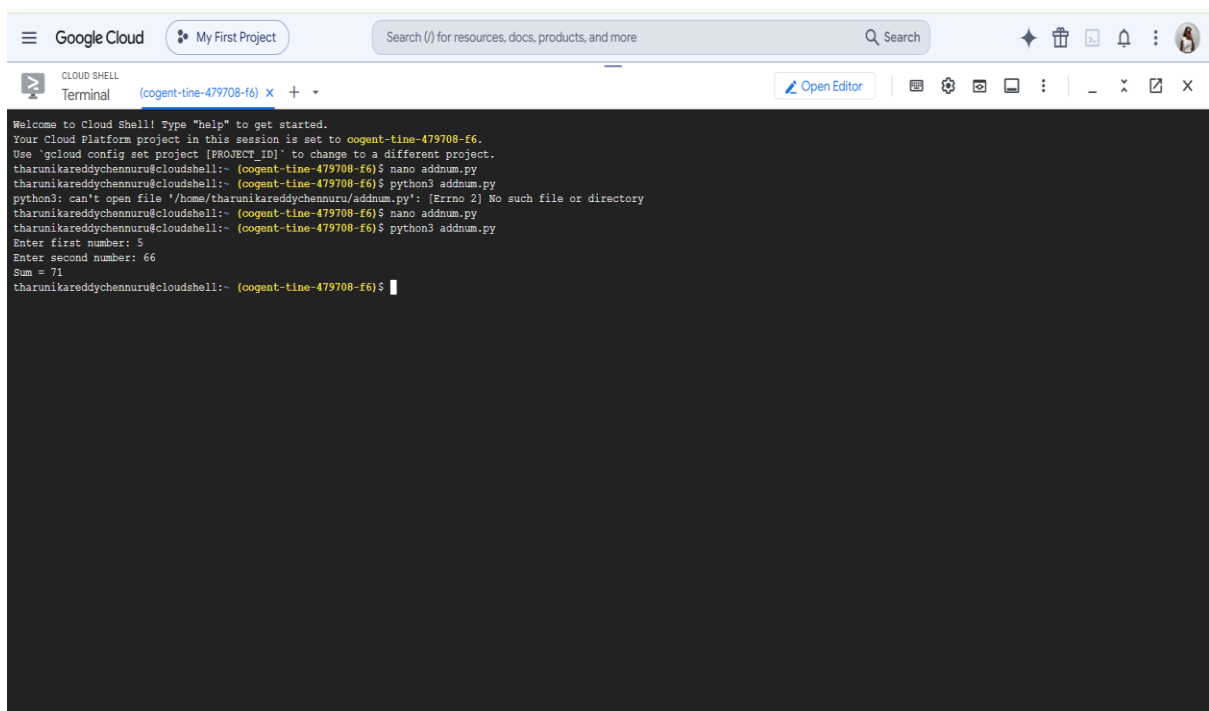
The screenshot shows the Google Cloud Shell interface. At the top, there's a header with the Google Cloud logo, 'My First Project', and a search bar. Below the header, the 'Terminal' tab is active, showing a nano editor editing a file named 'addnum.py'. The code in the editor is as follows:

```
GNU nano 1.2 addnum.py
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

sum = a + b

print("Sum =", sum)
```

The bottom of the editor shows a toolbar with various icons for file operations like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next, Back, and Forward.

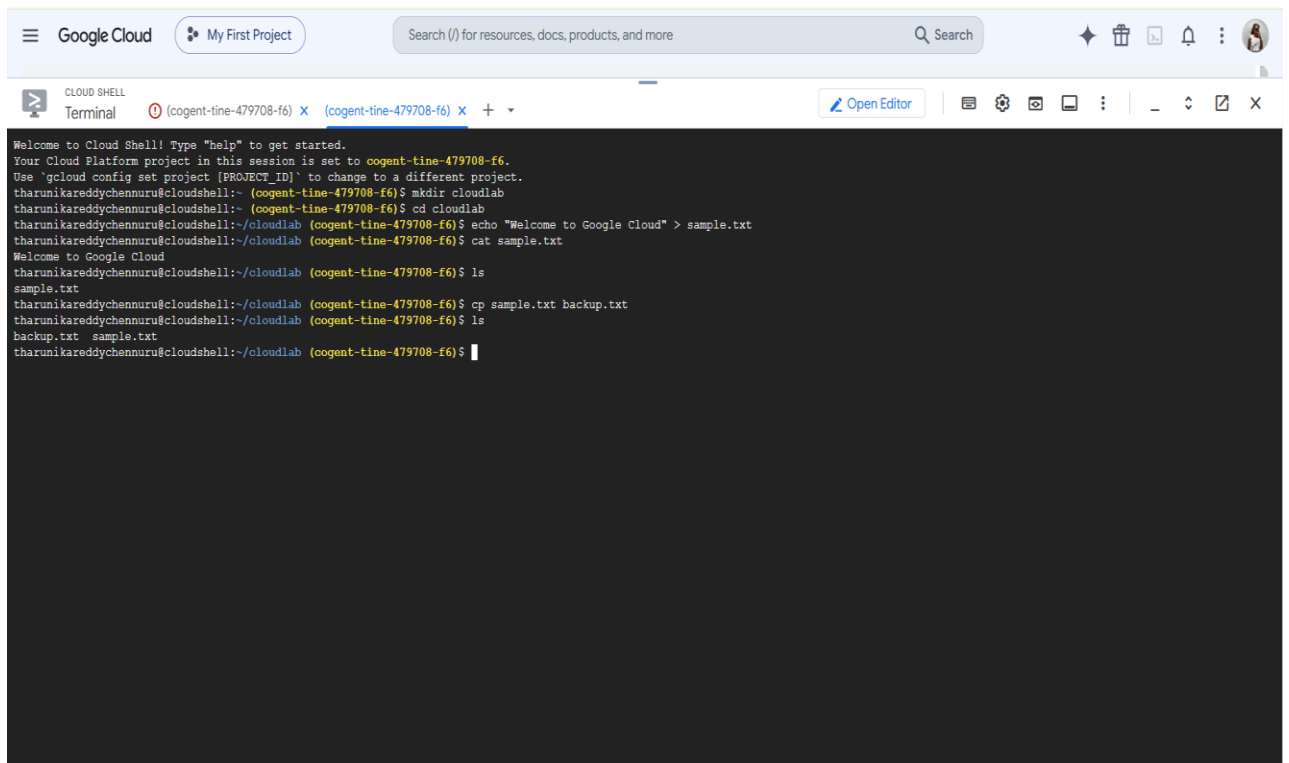


The screenshot shows the Google Cloud Shell interface with the 'Terminal' tab active. It displays the output of the previous actions:

```
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to coagent-tine-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddychnnuru@cloudshell:~ (coagent-tine-479708-f6)$ nano addnum.py
tharunikareddychnnuru@cloudshell:~ (coagent-tine-479708-f6)$ python3 addnum.py
python3: can't open file '/home/tharunikareddychnnuru/addnum.py': [Errno 2] No such file or directory
tharunikareddychnnuru@cloudshell:~ (coagent-tine-479708-f6)$ nano addnum.py
tharunikareddychnnuru@cloudshell:~ (coagent-tine-479708-f6)$ python3 addnum.py
Enter first number: 5
Enter second number: 66
Sum = 71
tharunikareddychnnuru@cloudshell:~ (coagent-tine-479708-f6)$
```

2. Create and Manage Files in Cloud Storage of Cloud Shell

Aim: Demonstrate cloud-based file creation and management.

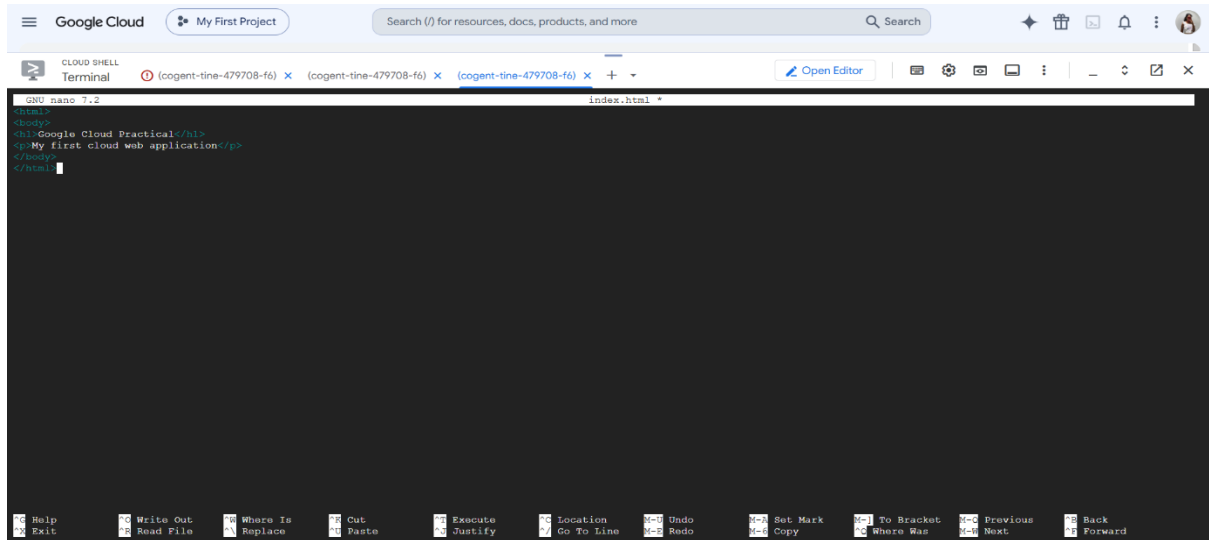


The screenshot displays the Google Cloud Shell interface. The top navigation bar includes the Google Cloud logo, a project selector set to 'My First Project', a search bar, and various utility icons. Below the navigation bar, the 'Terminal' tab is active, showing a series of commands and their outputs. The terminal session begins with a welcome message and instructions. The user then creates a directory named 'cloudlab', navigates into it, and creates a file named 'sample.txt' containing the text 'Welcome to Google Cloud'. Subsequently, the user lists the files, copies 'sample.txt' to 'backup.txt', and lists the files again to confirm the copy operation.

```
Cloud Shell
Terminal (cogent-tine-479708-f6) x (cogent-tine-479708-f6) + -
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to cogent-tine-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddyhennuru@cloudshell:~ (cogent-tine-479708-f6)$ mkdir cloudlab
tharunikareddyhennuru@cloudshell:~ (cogent-tine-479708-f6)$ cd cloudlab
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$ echo "Welcome to Google Cloud" > sample.txt
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$ cat sample.txt
Welcome to Google Cloud
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$ ls
sample.txt
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$ cp sample.txt backup.txt
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$ ls
backup.txt  sample.txt
tharunikareddyhennuru@cloudshell:~/cloudlab (cogent-tine-479708-f6)$
```

3. Run a Simple Web Server in Google Cloud Shell

Aim: Host and access a simple webpage from a cloud development environment.



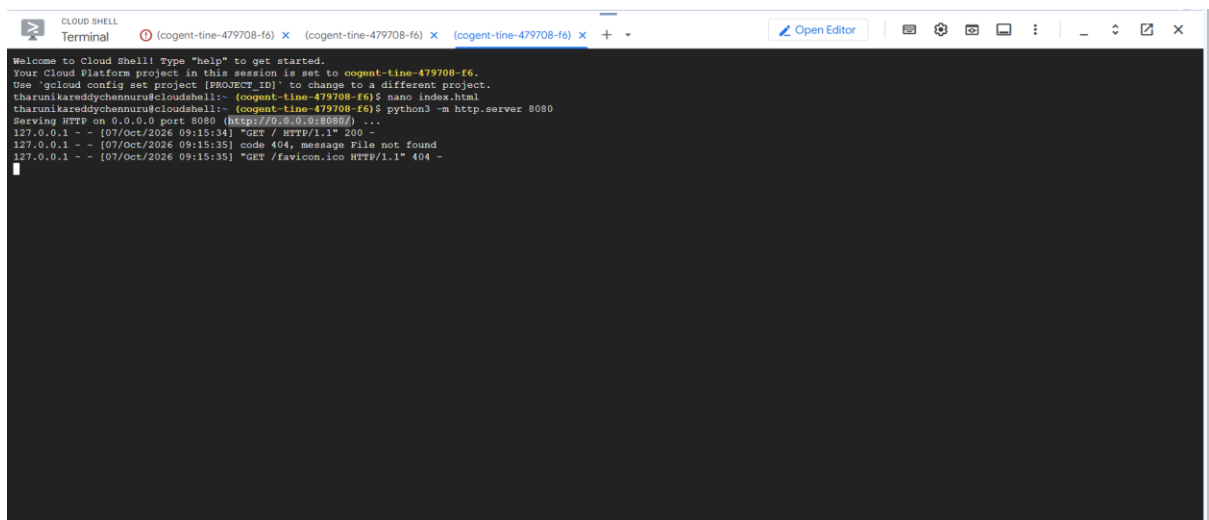
The screenshot shows the Google Cloud Shell interface. At the top, there's a header with 'Google Cloud', 'My First Project', and a search bar. Below the header, there's a tab bar with 'Terminal' and several instances of '(cogent-time-479708-f6)'. The main area is a terminal window running 'GNU nano 2.2'. The user has created a file named 'index.html' and entered the following HTML code:

```
<html>
<body>
<h1>Google Cloud Practical:</h1>
<p>My first cloud web application</p>
</body>
</html>
```

At the bottom of the terminal window, there's a toolbar with various icons and labels: Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next, Back, Forward.

Google Cloud Practical

My first cloud web application



The screenshot shows the Google Cloud Shell interface with the same header and tab bar as the previous image. The terminal window now shows the output of running 'python3 -m http.server 8080'. The output includes a welcome message, instructions on how to use 'gcloud config set project', and the command 'python3 -m http.server 8080'. Below this, there are several lines of HTTP request logs:

```
Starting HTTP on 0.0.0.0 port 8080 (https://0.0.0.0:8080/) ...
127.0.0.1 - - [07/Oct/2026 09:15:34] "GET / HTTP/1.1" 200 -
127.0.0.1 - - [07/Oct/2026 09:15:35] code 404, message File not found
127.0.0.1 - - [07/Oct/2026 09:15:35] "GET /favicon.ico HTTP/1.1" 404 -
```

4. System Resource Monitoring in Google Cloud Shell

Aim: Demonstrate monitoring of CPU, memory and storage resources in a cloud environment.

```
Google Cloud My First Project Search (/) for resources, docs, products, and more Q Search

CLOUD SHELL
Terminal (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x + Open Editor

Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to cogent-tine-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6)$ free -h
Mem: 7.8Gi total 62Gm used 6.0Gi free 1.1Mi shared 1.4Gi buff/cache 7.1Gi available
Swap: 0B total 0B used 0B free 0B shared 0B buff/cache 0B available
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6)$ df -h
Filesystem Size Used Avail Use% Mounted on
overlay 119G 108G 12G 91% /
tmpfs 64M 0 64M 0% /dev
/dev/sda1 119G 108G 12G 91% /root
/dev/disk/by-id/google-home-part1 4.9G 1.4M 4.6G 1% /home
shm 64M 0 64M 0% /dev/shm
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6)$ lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
Address sizes: 46 bits physical, 48 bits virtual
Byte Order: Little Endian
CPU(s): 2
On-line CPU(s) list: 0,1
Vendor ID: GenuineIntel
Model name: Intel(R) Xeon(R) CPU @ 2.20GHz
CPU family: 6
Model: 79
top - 09:18:34 up 1:49, 0 user, load average: 0.11, 0.04, 0.01
Tasks: 40 total, 1 running, 39 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.5 sy, 0.0 ni, 99.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7947.3 total, 6165.8 free, 638.6 used, 1384.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 7308.7 avail Mem

  PID USER      PR  NI   VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 776 tharuni+  10  -10   7623   3088   2004 S  0.0  0.0  0:00.13 tmux
    1 root      20   0   4328   3396   3112 S  0.0  0.0  0:00.08 bash
    9 syslog   20   0  222148 3912   2540 S  0.0  0.0  0:00.04 rsyslogd
   25 root      20   0  38804 28064 10832 S  0.0  0.3  0:01.67 python
   26 root      20   0   6204   2596   2352 S  0.0  0.0  0:00.00 lsofwr
   75 root     10 -10  12024 4016   2904 S  0.0  0.0  0:00.06 sshd
   79 root     10 -10  12020 2860   1756 S  0.0  0.0  0:00.00 sshd
```

```
Google Cloud My First Project Search (/) for resources, docs, products, and more Q Search

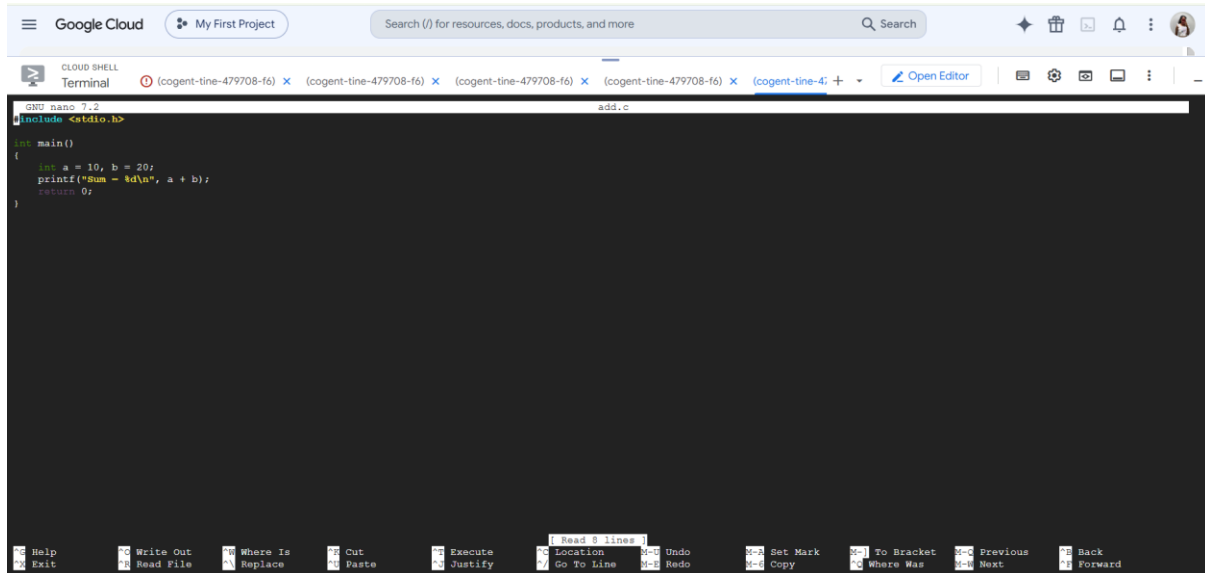
CLOUD SHELL
Terminal (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x (cogent-tine-479708-f6) x + Open Editor

Tasks: 40 total, 1 running, 39 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.5 us, 0.5 sy, 0.0 ni, 99.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7947.3 total, 6165.8 free, 638.6 used, 1384.1 buff/cache
MiB Swap: 0.0 total, 0.0 free, 0.0 used, 7308.7 avail Mem

  PID USER      PR  NI   VIRT  RES  SHR S %CPU  %MEM    TIME+  COMMAND
 776 tharuni+  10  -10   7623   3088   2004 S  0.0  0.0  0:00.13 tmux
    1 root      20   0   4328   3396   3112 S  0.0  0.0  0:00.08 bash
    9 syslog   20   0  222148 3912   2540 S  0.0  0.0  0:00.04 rsyslogd
   25 root      20   0  38804 28064 10832 S  0.0  0.3  0:01.67 python
   26 root      20   0   6204   2596   2352 S  0.0  0.0  0:00.00 lsofwr
   75 root     10 -10  12024 4016   2904 S  0.0  0.0  0:00.06 sshd
   79 root     10 -10  12020 2860   1756 S  0.0  0.0  0:00.00 sshd
  326 root      20   0 1267452 8332   6980 S  0.0  0.1  0:00.03 editor-proxy
  327 root      20   0  16136   6436   5512 S  0.0  0.1  0:00.01 sudo
  339 root      20   0 1227876 3044   2200 S  0.0  0.0  0:00.01 tmux-agent
  506 root      20   0 2027164 88716 64740 S  0.0  1.1  0:01.57 dockerd
  534 root      20   0 1747320 42452 30164 S  0.0  0.5  0:07.89 containerd
  747 root      20   0   2700   1648   1552 S  0.0  0.0  0:00.00 sleep
  777 tharuni+  10 -10   9320   6508   4112 S  0.0  0.1  0:00.06 bash
 1194 tharuni+ 10 -10   7732   4796   3132 S  0.0  0.1  0:00.00 nano
 1219 tharuni+ 10 -10   9320   6384   3980 S  0.0  0.1  0:00.08 bash
 1708 root     10 -10  13996   9700   8320 S  0.0  0.1  0:00.01 sshd
 1710 tharuni+ 10 -10  14256   6428   4768 S  0.0  0.1  0:00.01 sshd
 1711 tharuni+ 10 -10   7344   3764   3516 S  0.0  0.0  0:00.00 bash
 1712 tharuni+ 10 -10   7344   3772   3508 S  0.0  0.0  0:00.00 start-shell.sh
 1716 tharuni+ 10 -10   6880   3460   3160 S  0.0  0.0  0:00.00 tmux
 1717 tharuni+ 10 -10   9320   6520   4116 S  0.0  0.1  0:00.07 bash
 2102 root     10 -10  13996   9756   8380 S  0.0  0.1  0:00.01 sshd
 2104 tharuni+ 10 -10  14256   6400   4744 S  0.0  0.1  0:00.00 sshd
 2105 tharuni+ 10 -10   7344   3782   3540 S  0.0  0.0  0:00.00 bash
 2108 tharuni+ 10 -10   7608   4456   3856 S  0.0  0.1  0:00.00 bash
 2138 root     10 -10  13996   9732   8360 S  0.0  0.1  0:00.01 sshd
 2140 tharuni+ 10 -10  14256   6356   4708 S  0.0  0.1  0:00.00 sshd
 2141 tharuni+ 10 -10   7344   3782   3544 S  0.0  0.0  0:00.00 bash
 2142 tharuni+ 10 -10   7344   3782   3524 S  0.0  0.0  0:00.00 start-shell.sh
 2146 tharuni+ 10 -10   6880   3444   3148 S  0.0  0.0  0:00.00 tmux
 2147 tharuni+ 10 -10   9320   6560   4156 S  0.0  0.1  0:00.06 bash
tharunikareddychnnuru@cloudshell:~ (cogent-tine-479708-f6)$
```

5. Execute a C Program in Google Cloud Shell

Aim: Compile and execute a C application in a cloud-based environment.

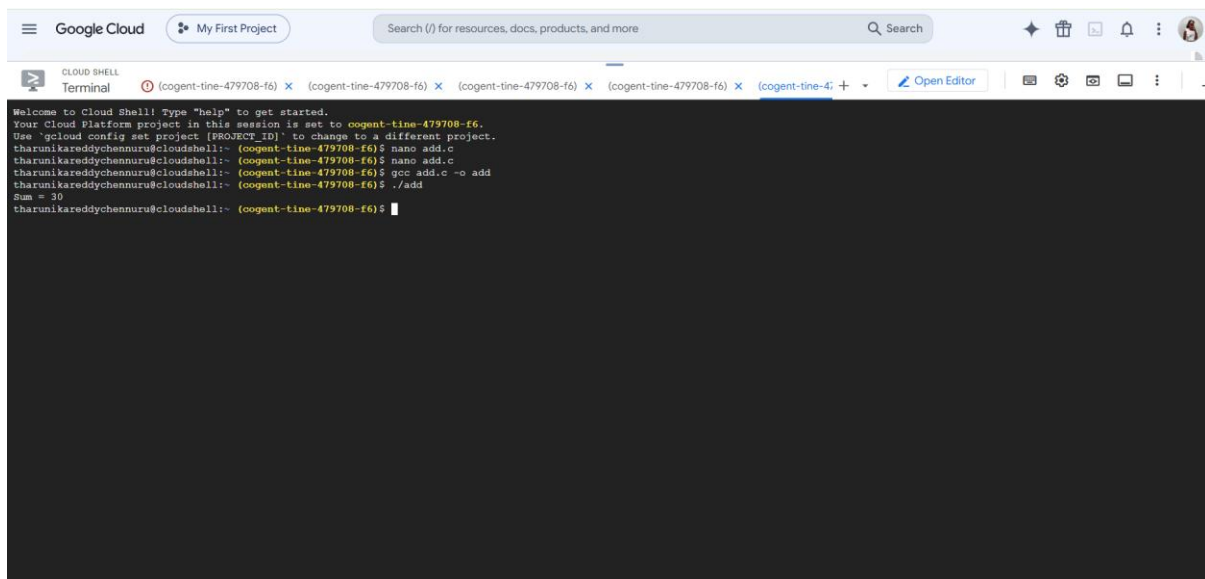


The screenshot shows the Google Cloud Shell interface with a code editor. The file being edited is named 'add.c'. The code is as follows:

```
GNU nano 2.2 add.c
#include <stdio.h>

int main()
{
    int a = 10, b = 20;
    printf("Sum = %d\n", a + b);
    return 0;
}
```

The bottom of the editor shows a toolbar with various editing tools like Help, Exit, Write Out, Read File, Where Is, Replace, Cut, Paste, Execute, Justify, Read 9 lines, Location, Go To Line, Undo, Redo, Set Mark, Copy, To Bracket, Where Was, Previous, Next, Back, and Forward.



The screenshot shows the Google Cloud Shell terminal window. It displays the following sequence of commands and output:

```
Welcome to Cloud Shell! Type "help" to get started.
Your Cloud Platform project in this session is set to cogent-time-479708-f6.
Use 'gcloud config set project [PROJECT_ID]' to change to a different project.
tharunikareddychnnuru@cloudshell:~$ nano add.c
tharunikareddychnnuru@cloudshell:~$ gcc add.c -o add
tharunikareddychnnuru@cloudshell:~$ ./add
Sum = 30
tharunikareddychnnuru@cloudshell:~$
```